

Functional Analysis

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1 Metric Spaces.

1.1 Topological Spaces

Definition 1.1. Given a set X , a topology τ on X is a subset of $\mathcal{P}(X)$ s.t.

1. $\emptyset, X \in \tau$
2. Closed under union: $E_\alpha \in \tau \forall \alpha \in I \implies \bigcup_{\alpha \in I} E_\alpha \in \tau$
3. Closed under finite intersection: $E_1, E_2, \dots, E_n \in \tau \implies \bigcap_{i=1}^n E_i \in \tau$

We say that (X, τ) is a topological space.

Definition 1.2. Let (X, τ) a topological space and $A \subseteq X$

- A open $\iff A \in \tau$.
- A is closed $\iff X \setminus A$ is open.

Definition 1.3. (X, τ) topological space and $x \in X$. A neighborhood of x is an open set $U_x \ni x$.

Definition 1.4. (X, τ) topological space, and $A \subseteq X$. Then x is an interior point of $A \iff \exists U_x$ an open neighborhood of x s.t. $U_x \subseteq A$.

We denote by $\mathring{A}$ the set of interior points of A . Notice $\mathring{A} \subseteq A$.

Now x is a limit point of $A \iff \forall U_x$ a neighborhood of x , we have that $U_x \setminus \{x\} \cap A \neq \emptyset$.
 A' denotes the set of all limit points of A and $\overline{A} := A \cup A'$ denotes the closure of A .

Proposition 1.5. 1. $\mathring{A}$ is an open set. In fact, A is open $\iff A = \mathring{A}$.

2. $\overline{A}$ is closed. In fact, A is closed $\iff A' \subseteq A \iff \overline{A} = A$.

Consequence, A is open iff $\forall x \in A, \exists U_x \ni x$ an open neighborhood s.t. $U_x \subseteq A$.

Proof. Easy. □

Definition 1.6. Let (X, τ_x) , (Y, τ_y) topological spaces, and $f: X \rightarrow Y$ and $x_0 \in X$. Then f is continuous at $x_0 \iff \forall V$ neighborhood of $f(x_0)$ there exists a neighborhood U of x_0 s.t. $f(U) \subseteq V$.

f is said to be continuous on X iff f is continuous at every $x_0 \in X$.

Proposition 1.7. $f: X \rightarrow Y$ continuous iff $f^{-1}(G)$ is open in X for every G open in Y .

Definition 1.8. (X, τ) topological spaces, and x_n a sequence in X . We say that x_n converges to $x_0 \in X$ iff for every U_{x_0} neighborhood of x_0 , $\exists N \in \mathbb{N}$ s.t. $x_n \in U_{x_0} \forall n \geq N$.

Example 1.9. $x_n = \frac{1}{n}$.

- In $(\mathbb{R}, \tau_{\text{trivial}})$, this sequence converges to every point in $\mathbb{R}$.
- In $(\mathbb{R}, \tau_{\text{disc}})$, this sequence doesn't converge to any point (we say that it diverges).

Proposition 1.10. (X, τ) topological space. $A \subseteq X$, x_n sequence in A that converges to x_0 . Then $x_0 \in \bar{A}$.

Proposition 1.11. $f: (X, \tau_x) \rightarrow (Y, \tau_y)$ and $x_0 \in X$ with f continuous at x_0 . If x_n converges to x_0 , then $f(x_n)$ converges to $f(x_0)$.

Definition 1.12. (X, τ) topological space, $A \subseteq X$. We say that A is dense in X iff $\bar{A} = X$. We say that X is separable iff it has a countable dense subset.

Example 1.13. $(\mathbb{R}^n, |\cdot|)$ is separable. $(\mathbb{R}, \tau_{\text{disc}})$ is not separable. $(X, \tau_{\text{trivial}})$ is always separable.

Definition 1.14. (X, τ) is a Hausdorff space iff $\forall x, y \in X$ $x \neq y \exists U_x, U_y$ neighborhoods of x, y respectively s.t. $U_x \cap U_y = \emptyset$.

Proposition 1.15. The limit of a convergent sequence in a Hausdorff space is unique.

Proof. Assume x_n converges to x, y with $x \neq y$, since Hausdorff we have U_x, U_y neighborhoods with $U_x \cap U_y = \emptyset$. Now for n big enough $x_n \in U_x$ for $n \geq N$ then $x_n \notin U_y$ for these $n \geq N$, contradiction. $\square$

Definition 1.16. X a set, then a metric on X is a function $d: X \times X \rightarrow \mathbb{R}$ s.t.

1. $d(x, y) \geq 0$, $\forall x, y \in X$ with equality iff $x = y$.
2. $d(x, y) = d(y, x) \forall x, y \in X$.
3. $d(x, z) \leq d(x, y) + d(y, z) \forall x, y, z \in X$

We say that (X, d) is a metric space.

Definition 1.17. $x \in X$ and $r \geq 0$ then

$$B(x, r) = \{y \in X: d(x, y) < r\}$$

Now define

$$\tau_d \{U \in \tau_d \iff \forall x \in U \exists r \geq 0 \text{ s.t. } B(x, r) \subseteq U\}$$

Proposition 1.18. *the $B(x, r)$. τ_d is a topology on X , called the metric topology. Then (X, τ_d) is always Hausdorff and τ_d is the smallest topology containing all*

Proposition 1.19. *Take (X, d) metric space and x_n converges to x_0 iff $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. $d(x_n, x_0) < \varepsilon \forall n \geq N$.*

Proposition 1.20. *(X, d) metric space and $E \subseteq X$, then $x \in \overline{E}$ iff $\exists x_n \in E$ a convergent sequence with $x_n \rightarrow x$*

Proof ($\Leftarrow$) Any topological space.

($\Rightarrow$) Take $x \in \overline{E} = E \cup E'$, if $x \in E$ take $x_n = x$. Now if $x \in E'$, then for every $n \in \mathbb{N}$, $\exists x_n \in B^*(x, 1/n) \cap E$ then $d(x_n, x) < 1/n$, by Archimedean property we get that $x_n \rightarrow x$. □

Proposition 1.21. *(X, d) metric space with $E \subseteq X$, then E is closed iff for every convergent sequence $x_n \in E$ the limit as $n \rightarrow \infty$ of $x_n \in E$.*

Definition 1.22. *(X, d) metric space, then $E \subseteq X$ is bounded iff $E \subseteq B(x_0, R)$ for some $x_0 \in x$ and $R > 0$. In particular, a sequence x_n is bounded iff $\exists M$ s.t. $|x_n| \leq M \forall n \in \mathbb{N}$.*

Example 1.23 (Some examples of some separable or not separable metric spaces).

- $(\mathbb{R}, |\cdot|)$ separable with countable dense set $\mathbb{Q}^n$.
- The space $C([a, b])$ of real valued continuous functions with $d_\infty(f, g) = \max_{[a, b]} |f(x) - g(x)|$. By Stone-Weierstrass, for every $f \in C([a, b])$ there exists a sequence of polynomials P_n s.t. $\lim_{n \rightarrow \infty} P_n = f$ uniformly. Let's call $P(\mathbb{R})$ the space of polynomials and fix $\varepsilon > 0$ and $f \in C([a, b])$. Then $\exists P \in P(\mathbb{R})$ s.t. $d_\infty(P, f) < \varepsilon/2$ with

$$p(x) = c_k x^k + c_{k-1} x^{k-1} + \cdots + c_1 x + c_0$$

Now for each $c_i \in q_i \in \mathbb{Q}$ s.t. $|c_i - q_i| < \frac{\varepsilon}{2M}$ where $M \geq \sum_{i=0}^k |x|^i, \forall x \in [a, b]$. Now let

$$q(x) = q_k x^k + \cdots + q_1 x + q_0$$

So we get that

$$\begin{aligned} |p(x) - q(x)| &\leq \sum_{i=0}^k \sum_{q_i - c_i} |x|^i \\ &\leq \frac{\varepsilon}{2M} \sum_{i=0}^k |x|^i \end{aligned}$$

So we get that

$$d_\infty(q, f) < \varepsilon$$

Hence $C([a, b])$ is separable.

- Let $\ell^\infty(\mathbb{R})$ be the space of bounded sequences. Now define

$$x = (x_1, x_2, \cdots), \quad y = (y_1, y_2, y_3, \cdots)$$

and

$$d_\infty(x, y) = \sup_{i \in \mathbb{N}} |x_i - y_i|$$

Assume it's separable, let M be the countable dense subset. Let $A = \{0, 1\}^{\mathbb{N}}$ be the space of all sequences with entries 0 or 1, clearly A is uncountable. But if $x, y \in A$ with $x \neq y$ then $d_\infty(x, y) = 1$. For every $a \in A$, construct $B(a, 1/2)$, let m_a be an element of $B(a, 1/2) \cap M$ (uses AOC and density of m_a). Clearly if $a \neq b$, we have $B(a, 1/2) \cap B(b, 1/2) = \emptyset$, then $a \rightarrow m_a$ is injective, hence M is uncountable. Contradiction.

Definition 1.24. Let (X, d) a metric space, given a sequence x_n in X , we say that x_n is Cauchy iff $\forall \varepsilon > 0, \exists N$ s.t.

$$d(x_n, x_m) < \varepsilon, \forall n, m \geq N$$

Proposition 1.25. Every Cauchy sequence is bounded, every convergent sequence is Cauchy.

Definition 1.26. (X, d) is complete iff every Cauchy Sequence converges.

Example 1.27. $(\mathbb{R}^n, |\cdot|)$ is complete.

Some properties of metric spaces.

- i) if x_n converges, then x_n is bounded.
- ii) Let C be the space of convergent sequences, then $C \subseteq \ell^\infty$. We want to show that C is closed in ℓ^∞ . Let $x^1 = (x_{1,1}, x_{1,2}, \dots), x^2 = (x_{2,1}, x_{2,2}, \dots), x^3 = (x_{3,1}, x_{3,2}, \dots), \dots$ be a sequence in C that converges to some sequence $x = (x_1, x_2, \dots) \in \ell^\infty$. Let $\varepsilon > 0$, since x^n converges to x in ℓ^∞ , $\exists N$ s.t $d_\infty(x^n, x) < \varepsilon/3 \forall n \geq N$. In particular for $n = N$ we have

$$|x_{N,i} - x_i| < \varepsilon/3, \quad \forall i = 1, 2, 3, \dots$$

Now x^N is Cauchy, then $\exists i_0$ s.t.

$$|x_{N,i} - x_{N,j}| < \varepsilon/3, \quad \forall i, j \geq i_0$$

Hence

$$\begin{aligned} |x_i - x_j| &\leq |x_i - x_{N,i}| + |x_{N,i} - x_{N,j}| + |x_{N,j} - x_j| \\ &< \varepsilon \end{aligned}$$

Then the sequence x is Cauchy, since $\mathbb{R}$ is complete we have that $x \in C$. So C is closed.

- iii) Is $\ell^\infty(\mathbb{R})$ complete. Let $x^1 = (x_{1,1}, x_{2,1}, \dots), x^2 = (x_{1,2}, x_{2,2}, \dots), \dots$ be a Cauchy sequence in $\ell^\infty(\mathbb{R})$. We define

$$x_i := (x_i^1, x_i^2, x_i^3, \dots)$$

Clearly we have that for each $i \in \mathbb{N}$,

$$d_\infty(x^k, x^l) \geq |x_{k,i} - x_{l,i}|$$

Hence it must be the case that all the x_i 's are Cauchy, and hence converge to some x_i^* (since $\mathbb{R}$ is complete). Define

$$x^* := (x_1^*, x_2^*, \dots)$$

Now by Cauchy we know that $\exists N \in \mathbb{N}$ s.t.

$$d_\infty(x_n, x_m) < \varepsilon/2, \quad \forall n, m \geq N$$

Therefore $\forall i \in \mathbb{N}$ we have that

$$|x_i^n - x_i^m| < \varepsilon/2$$

But we know that $(x_i^m)_i \rightarrow x_i^*$, by continuity of $|\cdot|$, we have

$$|x_i^n - x_i^*| \leq \varepsilon/2, \quad \forall i \in \mathbb{N}, \forall n \geq N$$

Note that N is fixed over all i 's hence

$$d_\infty(x^n, x^*) = \sup_{i \in \mathbb{N}} |x_i^n - x_i^*| \leq \varepsilon/2 < \varepsilon$$

Hence $\ell^\infty(\mathbb{R})$ is complete.

Note that x^* is bounded because each x^n is bounded in ℓ^∞ , therefore $|x_{n,i}| \leq M$, $\forall n, i$, taking $n \rightarrow \infty$ we get that $|x_i| \leq M$, $\forall i$.

- iv) Is $C([a, b])$ complete. This is just uniformly Cauchy iff uniformly convergent and the uniform limit of continuous functions is continuous.
- v) Is $P([a, b])$ closed in $C([a, b])$. No, this is because $P([a, b])$ is dense with nonempty complement. So if its complement was open, it should contain a nonempty open set, contradicting density of the polynomials.

Proposition 1.28. Let (X, d_x) , (Y, d_y) metric spaces and $f: X \rightarrow Y$ and $x_0 \in X$ then f is continuous at x_0 iff $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t.

$$d_y(f(x), f(x_0)) < \varepsilon, \quad x \text{ s.t. } d_x(x, x_0) < \delta$$

iff for every sequence $x_n \rightarrow x_0$ we have $f(x_n) \rightarrow f(x_0)$.

Proposition 1.29. (X, d) metric space, $E \subseteq X$. If E is complete, then E is closed.

Proof. Let x_n be a sequence in E that converges to some $x \in X$, but x_n is Cauchy in E so it converges to $x \in E$ (by uniqueness of the limit). $\square$

Example 1.30. Note that the converse is false, $(\mathbb{Q}, |\cdot|)$ is closed in itself but is not Cauchy.

Proposition 1.31. Every closed subset of a complete metric space is complete.

Proof. Take (X, d) complete and $E \subseteq X$ closed. Take a Cauchy sequence $x_n \in E$, since $E \subseteq X$ it converges to some $x \in X$. Now E is closed so we must have $x \in E$. $\square$

Example 1.32. $C \subseteq \ell^\infty(\mathbb{R})$ is complete.

Remark 1.33. (X, d) metric space:

$$|d(x, y) - d(x', y')| \leq d(x, x') + d(y, y')$$

This is because

$$\begin{aligned} d(x, y) &\leq d(x, x') + d(x', y') + d(y', y) \\ d(x, y) - d(x', y') &\leq d(x, x') + d(y, y') \end{aligned}$$

Similarly,

$$d(x', y) - d(x, y) \leq d(x', x) + d(y', y)$$

So we get the desired inequality.

Corollary 1.34. (X, d) metric space and x_n, y_n Cauchy sequences, then $d(x_n, y_n)$ converge in $\mathbb{R}$.

Proof. Take $d_n = d(x_n, y_n) \in \mathbb{R}$, we show that d_n is Cauchy. Now

$$\begin{aligned} |d_n - d_m| &= |d(x_n, y_n) - d(x_m, y_m)| \\ &\leq d(x_n, x_m) + d(y_n, y_m) \end{aligned}$$

Now since x_n, y_n are Cauchy then d_n is Cauchy in $\mathbb{R}$. Which by completeness gives that d_n converges. $\square$

Lemma 1.35. (X, d) metric space, and M a dense subset of X s.t. every Cauchy sequence in M converges in X . Then X is complete.

Proof. Let x_n be a Cauchy sequence in X , for every $n \in \mathbb{N}$, $\exists y_n \in M$ s.t.

$$d(x_n, y_n) < \frac{1}{n}$$

Let $\varepsilon > 0$, now

$$\begin{aligned} d(y_n, y_m) &\leq d(y_n, x_n) + d(x_n, x_m) + d(x_m, y_m) \\ &< \frac{1}{n} + d(x_n, x_m) + \frac{1}{m} \end{aligned}$$

Now $\exists N$ s.t. $n, m \geq N$ we have

$$d(y_n, y_m) < \varepsilon$$

This is because

$$1/n, 1/m < \varepsilon/3, \quad d(x_n, x_m) < \varepsilon/3$$

Now $y_n \in M$ is Cauchy, so y_n converges to some $y_0 \in X$. Now let ε

$$\begin{aligned} d(x_n, y_0) &\leq d(x_n, y_n) + d(y_n, y_0) \\ &< \frac{1}{n} + d(y_n, y_0) < \varepsilon \end{aligned}$$

for $n \geq N$ for some large enough index N . Then $x_n \rightarrow y_0$ so X is complete. $\square$

Definition 1.36. $\varphi: (X, d_X) \rightarrow (Y, d_Y)$ is an isometry iff

$$d_Y(\varphi(x), \varphi(y)) = d_X(x, y)$$

We say that A is isometric to B iff there exists a bijective isometry $\varphi: A \rightarrow B$.

Theorem 1.37 (Completeness theorem). (X, d) metric space, then there exists a complete metric space $(\hat{X}, \hat{d})$ and a dense subset M of $\hat{X}$ s.t. M is isometric to X .

In fact, the completion is "unique" (up to isometry), we call it the completion of X .

Proof. Let C_X be the space of Cauchy sequences in X . We define the relation $\sim$ on C_X as follows

$$x_n \sim y_n \iff \lim_{n \rightarrow \infty} d(x_n, y_n) = 0$$

Not that the RHS exists by the above. Clearly $\sim$ is an equivalence relation. Let $\hat{X} = C_X / \sim$ (the set of equivalence classes), i.e.

$$\hat{X} = \{[x_n] : x_n \in C_X\}$$

We define $\hat{d}([x_n], [y_n]) = \lim_{n \rightarrow \infty} d(x_n, y_n)$ (this exists by the corollary above). Note that this is well-defined, indeed let $x_n \sim x'_n$ and $y_n \sim y'_n$ so we have that

$$|d(x_n, y_n) - d(x'_n, y'_n)| \leq d(x_n, x'_n) + d(y_n, y'_n) \rightarrow 0$$

Therefore

$$\lim_{n \rightarrow \infty} d(x_n, y_n) = \lim_{n \rightarrow \infty} d(x'_n, y'_n)$$

We now show that $\hat{d}$ is a metric (clear since d is a metric):

- Clearly $\hat{d}([x_n], [y_n]) \geq 0$ and equal to zero iff $\lim_{n \rightarrow \infty} d(x_n, y_n) = 0 \iff x_n \sim y_n \iff [x_n] = [y_n]$.
- Symmetry follows from Symmetry of d .
- Triangle inequality follows from triangle inequality on d .

Define $\varphi: X \rightarrow \hat{X}$ as follows,

$$\varphi(x) = [(x, x, x, \dots)], \quad (x, x, x, \dots) \in C_x$$

This is clearly an isometry, we denote by $M := \varphi(X) = \{[(x, x, x, \dots)] : x \in X\}$. We now show that $\overline{M} = \hat{X}$. Consider an element $[x_n] = [(x_1, x_2, \dots)] \in \hat{X}$. Let $\varepsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $\forall m, n \geq N$

$$d(x_n, x_m) < \varepsilon/2$$

Consider the sequence $P_n = (x_N, x_N, \dots)$, clearly

$$d(x_n, P_n) = d(x_n, x_N) < \varepsilon/2, \quad \forall n \geq N$$

Then

$$\hat{d}([x_n], [P_n]) = \lim_{n \rightarrow \infty} d(x_n, P_n) \leq \varepsilon/2 < \varepsilon$$

With

$$[P_n] = [(x_N, x_N, \dots)] \in M$$

Therefore $\overline{M} = \hat{X}$. It remains to show that $\hat{X}$ is complete, we use the lemma. Take a Cauchy sequence in M ,

$$P_{x_1} = [(x_1, x_1, \dots)], P_{x_2} = [(x_2, x_2, \dots)], \dots, P_{x_k} = [(x_k, x_k, x_k, \dots)]$$

Construct the sequence $p = (x_1, x_2, x_3, \dots)$, note that

$$d(x_n, x_m) = \hat{d}([(x_n)], [(x_m)]) = \hat{d}(P_{x_n}, P_{x_m})$$

But P_{x_n} is Cauchy in $\hat{X}$, hence p is Cauchy in X , so $P := [p] \in \hat{X}$. Take $\varepsilon > 0$, then $\exists N \in \mathbb{N}$ s.t.

$$d(x_n, x_k) \leq \varepsilon/2, \quad \forall n, k \geq N$$

So we get that for $k \geq N$

$$\begin{aligned} \hat{d}(P, P_{x_k}) &= \lim_{n \rightarrow \infty} d(x_n, x_k) \\ &\leq \varepsilon/2 \\ &< \varepsilon \end{aligned}$$

Hence

$$\lim_{k \rightarrow \infty} P_{x_k} = P \in \hat{X}$$

□

1.2 Compactness

Definition 1.38. (X, τ) topological spaces and $K \subseteq X$. Then K is compact iff every open cover $\mathcal{G} = \{G_\alpha\}_{\alpha \in I}$ of K has a finite subcover.

Example 1.39. $[0, 1]$ is not compact in $(\mathbb{R}, |\cdot|)$, $\mathcal{G} = \{(-1, 1 - 1/n) : n \in \mathbb{N}\}$ has no finite subcover. Finite sets are clearly compact.

Proposition 1.40. (X, τ) topological space, then every closed subset of a compact set is compact.

Proof. Let K be compact, and $E \subseteq K$ is closed. Let $\mathcal{G} = \{G_\alpha\}_{\alpha \in I}$ open cover of E , so

$$E \subseteq \bigcup_{\alpha \in I} G_\alpha$$

But E^c is open with

$$K \subseteq \left(\bigcup_{\alpha \in I} G_\alpha \right) \cup E^c$$

Hence $\{G_\alpha\} \cup \{E^c\}$ is an open cover of K , so it has a finite subcover

$$K \subseteq G_{\alpha_1} \cup \dots \cup G_{\alpha_N} \cup E^c$$

Then clearly

$$E = E \cap K \subseteq G_{\alpha_1} \cup \dots \cup G_{\alpha_N}$$

so E is compact. □

Proposition 1.41. *Let (X, τ) be a Hausdorff space, then K compact $\implies K$ is closed.*

Proof. Take $q \in K^c$, for every $p \in K$ by Hausdorff $\exists U_p, V_p^q$ open neighborhoods of p, q resp. s.t. $U_p \cap V_p^q = \emptyset$. Take $\mathcal{G} = \{U_p\}_{p \in K}$ an open cover of K , so it has a finite subcover

$$K \subseteq U_{p_1} \cup \dots \cup U_{p_N}$$

Now define

$$V^q = V_{p_1}^q \cap \dots \cap V_{p_N}^q$$

open set with

$$V^q \cap K = \emptyset$$

So K is closed. □

Proposition 1.42. *(X, d) metric space, $K \subseteq X$ is compact $\implies K$ is closed and bounded.*

Proof. Ex. □